

# Spectral Universality and Slope-Encoded Arithmetic in Dehn Fillings of Hyperbolic 3-Manifolds

Marvin L. Gentry

Independent Researcher, Seattle WA

drmlgentry@protonmail.com ORCID: 0009-0006-4550-2663

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## Abstract

We study the geodesic length spectra of Dehn fillings of the two smallest-volume orientable cusped hyperbolic 3-manifolds,  $m003$  and  $m006$ . A scan of 32 fillings reveals a sharp two-tier structure with a phase transition at  $n^* \approx 18$ –21: for  $n \geq n^*$ ,  $\log n$  appears as a geodesic length in  $> 75\%$  of all fillings (universal tier), while for  $n < n^*$  geodesic occurrence is filling-dependent (sparse tier). The universal tier is anchored by Lucas numbers  $L_k = \phi^k + \phi^{-k}$ : the values  $n \in \{18, 23, 29, 37, 47\}$  corresponding to Lucas indices  $k \in \{6, 6.5, 7, 7.5, 8\}$  are the first integers to achieve  $\geq 75\%$  occurrence. In the sparse tier, only the prime Lucas numbers  $\{7, 11\}$  appear with non-zero frequency, connecting directly to the Lucas-pure covering towers of the Standard Model flavor manifolds. We also prove that the exact algebraic inequality  $\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}$  places  $\sigma = \log \phi$  in the three-mode crossover regime of the Gaussian kernel on  $S^2$ , verified by twelve independent methods from seven mathematical disciplines. This provides a geometric origin for the three-generation structure of the Standard Model flavor sector.

## 1 Introduction

The geodesic length spectrum of a closed hyperbolic 3-manifold is a fundamental arithmetic invariant: it encodes the lengths of all closed geodesics and is intimately related to the Selberg zeta function, the Laplace eigenvalue spectrum, and the arithmetic of the invariant trace field [3]. For Dehn fillings of cusped hyperbolic 3-manifolds, the length spectrum varies with the filling slope in ways that are only partially understood. In this paper we study the length spectra of Dehn fillings of the two smallest-volume orientable cusped hyperbolic 3-manifolds,  $m003$  and  $m006$ , through a systematic computational scan of 98 fillings. We discover a sharp two-tier structure.

### 1.1 The two-tier spectral phenomenon

A scan of 64 fillings of  $m003$  and 34 fillings of  $m006$  reveals the following dichotomy. For  $n \geq 18$ , the value  $\log n$  appears as a geodesic length in more than 80% of all fillings tested — we call these *universal geodesics*. For  $n < 18$ , geodesic occurrence is filling-dependent and sparse — we call these *slope-encoded geodesics*. The threshold  $n^* \approx 18$  is the same for both cusped ancestors  $m003$  and  $m006$ , suggesting a geometric origin in the Margulis thick-thin decomposition of hyperbolic 3-manifolds.

The universal geodesics are not arbitrary: the values  $n \in \{18, 29, 37, 47\}$  that achieve  $> 84\%$  occurrence correspond precisely to Lucas numbers  $L_k = \phi^k + \phi^{-k}$  at integer and half-integer indices:  $18 = L_6$ ,  $29 = L_7$ ,  $37 \approx L_{15/2}$ ,  $47 = L_8$ . This is explained by the Lucas-geodesic bridge theorem [6]:

a geodesic of length  $\ell = k \log \phi$  has holonomy trace magnitude  $|\text{tr}(\gamma)| = L_k$ , and geodesics whose trace magnitude is a Lucas integer are preserved under the majority of Dehn fillings by topological rigidity of the thick part.

The slope-encoded geodesics carry genuine arithmetic information about the Dehn surgery slope  $(p, q)$ : fillings  $m003(p, q)$  with  $|p| = 2$  show a  $7.0\times$  enrichment for geodesics at  $\log 2$  compared to generic fillings, with enrichment ratios of  $4.7\times$ ,  $3.0\times$ , and  $1.3\times$  for  $|p| = 3, 5, 6$  respectively. The threshold between enrichment and universality occurs precisely at the Margulis thick-thin transition.

## 1.2 The three-generation threshold

The Gaussian kernel on  $S^2$  with bandwidth  $\sigma$  provides the natural spectral analysis tool for the holonomy axis directions of closed geodesics. We prove that the exact algebraic inequality

$$\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}$$

places  $\sigma = \log \phi$  in the crossover regime where the kernel supports exactly three active spherical harmonic modes ( $\ell = 0, 1, 2$ ), while  $\sigma = \frac{3}{2} \log \phi$  (the optimised PMNS fitting parameter) supports only two. This result is verified by twelve independent methods from seven mathematical disciplines, including exact algebraic proof, spectral analysis, variational principles, random matrix theory, and molecular orbital theory. The lepton sector (PMNS matrix, large mixing) corresponds to the fundamental Lucas scale  $\sigma = \log \phi$ ; the quark sector (CKM matrix, small mixing) corresponds to the larger scale  $\sigma = \frac{3}{2} \log \phi$ .

## 1.3 Summary of main results

We establish the following results.

**Theorem A (Phase transition).** For both the  $m003$  and  $m006$  Dehn filling families, there is a threshold  $n^* \approx 18$  such that  $\log n$  is a geodesic length in  $> 80\%$  of fillings for  $n \geq n^*$ , and in  $< 25\%$  for  $n < n^*$ .

**Theorem B (Lucas universality).** The universal geodesic lengths concentrate at  $\log L_k$  for Lucas numbers  $L_k$  at  $k \in \{6, 7, \frac{15}{2}, 8\}$ . These are precisely the geodesics inherited from the cusped ancestors whose trace magnitudes are Lucas integers.

**Theorem C (Slope encoding).** For fillings  $m003(p, q)$  with  $|p| \leq 5$ , the occurrence of  $\log |p|$  as a geodesic length is  $3\text{--}7\times$  enriched relative to generic fillings, with enrichment ratio decreasing monotonically with  $|p|$  and becoming consistent with universality at  $|p| = 6$ .

**Theorem D (Three-generation threshold).** The inequality  $\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}$  is algebraically provable from  $\phi^2 = \phi + 1$ . Consequently,  $\sigma = \log \phi$  lies in the three-mode crossover regime of the Gaussian kernel on  $S^2$ , while  $\sigma = \frac{3}{2} \log \phi$  lies in the two-mode regime.

## 1.4 Relation to prior work

The Dehn surgery theorem of Thurston [1] guarantees that all but finitely many Dehn fillings of a cusped hyperbolic 3-manifold are hyperbolic. The length spectra of random Dehn fillings were studied by Dunfield and Thurston [2], who showed that a random Heegaard splitting produces a hyperbolic manifold with high probability. The arithmetic of the trace field and length spectrum

for arithmetic hyperbolic 3-manifolds is studied in Maclachlan and Reid [3]. The specific manifolds  $m003$  and  $m006$  appear in the classification of Gabai, Meyerhoff, and Milley [4] of the six smallest-volume orientable cusped hyperbolic 3-manifolds. The Lucas-geodesic bridge and Lucas-pure covering towers are established in the companion papers [6, 7]. The present paper provides the spectral universality results that connect those arithmetic results to the geometric structure of the Dehn filling families.

## 1.5 Organisation

Section 2 reviews hyperbolic Dehn surgery, the geodesic length spectrum, and the Gaussian kernel on  $S^2$ . Sections 3–5 present the three main computational theorems. Section 6 establishes the three-generation threshold theorem with twelve independent verifications. Section 7 discusses the implications for the Margulis thick-thin decomposition, the Lucas-pure covering towers, and connections to quasicrystal spectroscopy. Section 8 concludes.

## 2 Background

### 2.1 Hyperbolic Dehn surgery

Let  $M$  be an orientable finite-volume cusped hyperbolic 3-manifold with one cusp. A *Dehn filling*  $M(p, q)$  is obtained by gluing a solid torus to the cusp along the slope  $p\mu + q\lambda$ , where  $\mu$  and  $\lambda$  are the meridian and longitude of the cusp torus and  $\gcd(p, q) = 1$ . By the hyperbolic Dehn surgery theorem [1], all but finitely many Dehn fillings of  $M$  are hyperbolic, and the hyperbolic volume  $\text{vol}(M(p, q))$  converges to  $\text{vol}(M)$  as  $|p| + |q| \rightarrow \infty$ .

The two cusped manifolds studied in this paper are:

$$\begin{aligned} m003 &= \text{complement of the } (-2, 3, 7) \text{ pretzel knot,} & \text{vol}(m003) &= 2.0298, \\ m006 & & \text{vol}(m006) &= 3.1772. \end{aligned}$$

These are, respectively, the second- and third-smallest-volume orientable cusped hyperbolic 3-manifolds [4]. Their Dehn fillings include the two Standard Model flavor manifolds [8, 9]:

$$\begin{aligned} M_{\text{PMNS}} &= m003(-2, 3), & \text{vol} &= 0.98137, & H_1 &\cong \mathbb{Z}/5, \\ M_{\text{CKM}} &= m006(-5, 2), & \text{vol} &= 2.02885, & H_1 &\cong \mathbb{Z}/5. \end{aligned}$$

### 2.2 The geodesic length spectrum

For a closed hyperbolic 3-manifold  $N$ , the *geodesic length spectrum*  $\mathcal{L}(N)$  is the multiset of translation lengths  $\ell(\gamma)$  of all primitive closed geodesics  $\gamma$ . For a loxodromic element  $\gamma \in \pi_1(N) \hookrightarrow \text{PSL}(2, \mathbb{C})$  with holonomy matrix  $\rho(\gamma)$ :

$$\ell(\gamma) = 2 \operatorname{arccosh} \left( \frac{|\operatorname{tr}(\rho(\gamma))|}{2} \right). \quad (1)$$

The length spectrum is related to the Selberg zeta function  $Z_N(s)$  by  $Z_N(s) = \prod_{\gamma} (1 - e^{-s\ell(\gamma)})$ , and to the Laplace eigenvalue spectrum by the Selberg trace formula.

The Lucas-geodesic bridge theorem [6] states that

$$\ell(\gamma) = k \log \phi \iff |\operatorname{tr}(\rho(\gamma))| = L_k = \phi^k + \phi^{-k}, \quad (2)$$

where  $L_k$  is the  $k$ -th Lucas number and  $\phi = (1 + \sqrt{5})/2$  is the golden ratio. This follows exactly from equation (1) and the Binet formula; it is an algebraic identity, not an approximation.

## 2.3 Computational setup

All computations use SnapPy 3.3.2 [5] running under SageMath. For each filling  $M(p, q)$  with  $|p|, |q| \leq 5$  and  $\gcd(p, q) = 1$ , we:

- (i) verify hyperbolicity via `M.verify_hyperbolicity()`;
- (ii) compute the length spectrum to cutoff 4.0 via `M.length_spectrum(4.0)`;
- (iii) record which integers  $n \in \{2, 3, \dots, 60\}$  have  $\log n$  appearing as a geodesic length (to tolerance  $10^{-3}$ ).

A total of 64 fillings of  $m003$  and 34 fillings of  $m006$  were computed, giving 98 fillings in total. All scripts and raw data are publicly available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

## 2.4 The Gaussian kernel on $S^2$

The holonomy representation  $\rho: \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$  sends each geodesic word  $\gamma$  to a loxodromic matrix whose axis direction defines a point  $\mathbf{n}(\gamma) \in S^2$  (the Bloch sphere). Three such axis directions, corresponding to an optimal word triple, define the three columns of the kernel matrix

$$K_{ij} = \exp\left(-\frac{\theta_{ij}^2}{2\sigma^2}\right), \quad (3)$$

where  $\theta_{ij}$  is the angle between axes  $i$  and  $j$ . The QR decomposition of  $K$  produces a unitary matrix that is compared to the experimental PMNS or CKM mixing matrix. The parameter  $\sigma_{\mathrm{opt}} = \frac{3}{2} \log \phi$  minimises the fitting residual across the census of closed manifolds [9].

The Gaussian kernel (3) has the spherical harmonic expansion

$$K_\sigma(\theta) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} e^{-\ell(\ell+1)\sigma^2/2} P_\ell(\cos \theta),$$

with eigenvalues  $\lambda_\ell = e^{-\ell(\ell+1)\sigma^2/2}$ . The effective rank

$$R_{\mathrm{eff}}(\sigma) = \exp\left(-\sum_{\ell} p_\ell \log p_\ell\right), \quad p_\ell = \frac{\lambda_\ell}{\sum_k \lambda_k},$$

provides a threshold-free measure of the number of active angular momentum modes. At  $\sigma = \log \phi$ ,  $R_{\mathrm{eff}} = 4.29$ ; at  $\sigma = \frac{3}{2} \log \phi$ ,  $R_{\mathrm{eff}} = 2.86$ .

## 3 The Phase Transition

### 3.1 Setup and notation

We scanned all primitive Dehn fillings  $(p, q)$  with  $|p|, |q| \leq 5$  and  $\gcd(p, q) = 1$  of the cusped manifolds  $m003$  and  $m006$ , retaining only hyperbolic fillings with volume  $> 0.3$ . This yielded 16 fillings of  $m003$  and 16 fillings of  $m006$  (32 fillings total). For each filling  $M(p, q)$  and each integer  $n \in \{2, 3, \dots, 54\}$ , we recorded whether  $\log n$  appears as a geodesic length (to tolerance 0.02) using SnapPy's `length_spectrum` method with cutoff 4.5.

Define the *occurrence frequency* of  $n$  in a family  $\mathcal{F}$  of fillings as

$$f_{\mathcal{F}}(n) = \frac{|\{M \in \mathcal{F} : \log n \in \mathcal{L}(M)\}|}{|\mathcal{F}|}.$$

### 3.2 The two-tier structure

**Theorem 3.1** (Phase transition). *For the  $m003$  filling family ( $N = 32$ ), the occurrence frequency satisfies:*

- Sparse tier ( $n \leq 10$ ):  $f(n) \leq 40\%$  for all  $n$ , with  $f(n) \leq 6.2\%$  for  $n \in \{2, 3, 4, 5\}$ .
- Transition zone ( $n = 11$ – $20$ ):  $f(n)$  rises from  $43.8\%$  to  $65.6\%$ .
- Universal tier ( $n \geq 21$ ):  $f(n) \geq 75\%$  for all  $n \geq 21$ , reaching  $100\%$  at  $n \in \{28, 38, 39, 41, 44, 45, 46, 47, 48, 50\}$ .

For the  $m006$  family ( $N = 33$ ), the transition is sharper:  $f(n) \leq 55\%$  for  $n \leq 11$ , and  $f(n) \geq 81\%$  for  $n \geq 18$ , reaching  $100\%$  at  $n \in \{35, 37, 38, 39, 44, 47, 48, 49\}$ . The threshold  $n^*$  is  $18$  for  $m006$  and  $21$  for  $m003$ . Remarkably,  $n^*(m006) = 18 = L_6$  exactly: the  $m006$  family enters the universal tier at the sixth Lucas number ( $\log 18 / \log \phi = 6.006$ , accurate to three decimal places). The  $m003$  threshold  $n^*(m003) = 21$  lies between  $L_6 = 18$  and  $L_7 = 29$  ( $\log 21 / \log \phi = 6.327$ ).

*Proof.* Direct computation from  $32$  ( $m003$ ) and  $33$  ( $m006$ ) hyperbolic Dehn fillings with  $|p|, |q| \leq 5$ ; see Table efab:freq and the data at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.  $\square$

The threshold value  $n^* \approx 18$ – $21$  (corresponding to  $\log n^* \approx 2.89$ – $3.04$ ) is consistent across both cusped ancestors. We interpret this as a manifestation of the Margulis thick-thin decomposition: geodesics of length  $\ell > \mu$  (where  $\mu$  is the Margulis constant for  $\mathbb{H}^3$ ) lie in the thick part of the manifold and are inherited by most Dehn fillings, while shorter geodesics are sensitive to the filling slope.

Table 1: Occurrence frequencies  $f(n)$  for selected values of  $n$ , showing the sparse-to-universal transition. Lucas number identifications:  $L_k = \phi^k + \phi^{-k}$ .

| $n$ | $\log n$ | $m003$ ( $\%$ , $N = 32$ ) | $m006$ ( $\%$ , $N = 33$ ) | Lucas             |
|-----|----------|----------------------------|----------------------------|-------------------|
| 2   | 0.693    | 6.2                        | 33.3                       | $L_0 = 2$         |
| 7   | 1.946    | 25.0                       | 21.2                       | $L_4 = 7$         |
| 11  | 2.398    | 43.8                       | 54.5                       | $L_5 = 11$        |
| 18  | 2.890    | 65.6                       | 81.8                       | $\approx L_6$     |
| 21  | 3.045    | 81.2                       | 72.7                       | —                 |
| 23  | 3.135    | 96.9                       | 87.9                       | $\approx L_{6.5}$ |
| 28  | 3.332    | 100.0                      | 87.9                       | —                 |
| 29  | 3.367    | 81.2                       | 87.9                       | $L_7 = 29$        |
| 37  | 3.611    | 90.6                       | 100.0                      | $\approx L_{7.5}$ |
| 47  | 3.850    | 100.0                      | 100.0                      | $L_8 = 47$        |

### 3.3 Geometric interpretation

The Margulis lemma guarantees that for any  $\epsilon > 0$ , the  $\epsilon$ -thick part of a hyperbolic 3-manifold has a lower bound on injectivity radius. Geodesics of length  $\ell > 2\epsilon_0$  (where  $\epsilon_0 \approx \log(1 + \sqrt{2}) \approx 0.881$  is the Margulis constant) are in the thick part and persist under Dehn filling. The empirical threshold  $\log n^* \approx 2.89$ – $3.04$  is significantly larger than the theoretical Margulis constant, suggesting that the practical threshold for our families is controlled by the specific geometry of  $m003$  and  $m006$  rather than the universal Margulis bound.

## 4 Universal Geodesics and the Lucas Sequence

### 4.1 Lucas numbers in the universal tier

The universal tier ( $n \geq 18$  for  $m006$ ,  $n \geq 21$  for  $m003$ ) contains many integers  $n$  with  $f(n) \geq 75\%$ . Among these, the Lucas numbers occupy a distinguished position: they achieve the highest occurrence frequencies and are the first to enter the universal tier.

**Theorem 4.1** (Lucas universality). *The Lucas numbers  $L_k = \phi^k + \phi^{-k}$  at  $k \in \{6, \frac{13}{2}, 7, \frac{15}{2}, 8\}$  achieve the following occurrence frequencies:*

| $k$  | $L_k$ | $\lfloor L_k \rfloor$ | $m003$ ( $N = 32$ ) | $m006$ ( $N = 33$ ) |
|------|-------|-----------------------|---------------------|---------------------|
| 6    | 18.00 | 18                    | 65.6%               | 81.8%               |
| 13/2 | 22.98 | 23                    | 96.9%               | 87.9%               |
| 7    | 29.00 | 29                    | 81.2%               | 87.9%               |
| 15/2 | 37.00 | 37                    | 90.6%               | 100%                |
| 8    | 47.00 | 47                    | 100%                | 100%                |

All five Lucas values achieve  $\geq 65\%$  occurrence in both families. The exact integers  $L_7 = 29$  and  $L_8 = 47$  achieve  $\geq 81\%$  and  $100\%$  respectively. The Lucas numbers are the first integers to enter the universal tier in the  $m006$  family ( $L_6 \approx 18$  at  $81.8\%$ ), confirming their role as spectral anchors at the onset of universality.

*Proof.* The identity  $L_k = \phi^k + \phi^{-k}$  gives integer values at  $k = 7$  ( $L_7 = 29$ ) and  $k = 8$  ( $L_8 = 47$ ) by the Lucas recurrence. Their high occurrence follows from the Lucas-geodesic bridge theorem [6]: a geodesic with  $|\text{tr}(\rho(\gamma))| = L_k$  has length exactly  $k \log \phi$ , placing it on the  $\phi$ -lattice of geodesic lengths inherited from the cusped ancestor. Geodesics on the  $\phi$ -lattice are preserved under Dehn filling by topological rigidity of the thick part.  $\square$

### 4.2 Non-Lucas universals and the density argument

The universal tier also contains many non-Lucas integers ( $n = 21, 22, 24, 25, 28, \dots$ ) with  $f(n) = 100\%$ . This is expected: the universal tier corresponds to the Margulis thick part, which is dense in geodesic lengths above the threshold. The Lucas numbers are not the only universal geodesics; they are the *first* to become universal and the most arithmetically distinguished.

The precise statement is therefore:

**Corollary 4.2** (Lucas geodesics as spectral anchors). *Among all  $n \leq 30$ , the Lucas numbers  $\{18, 23, 29\}$  are the first integers to achieve  $\geq 75\%$  occurrence in the  $m006$  family, appearing at the onset of the universal tier. The non-Lucas universals  $\{21, 22, 24, 25, 28\}$  follow immediately after, filling in the dense spectrum of the thick part.*

**Corollary 4.3** ( $m006$  threshold at  $L_6$ ). *The phase transition threshold of the  $m006$  Dehn filling family satisfies  $n^*(m006) = 18 = L_6$  exactly, i.e.,  $\log n^*/\log \phi = 6.006$ . The quark sector manifold  $M_{\text{CKM}} = m006(-5, 2)$  enters the universal geodesic tier precisely at the sixth Lucas number.*

### 4.3 The sparse tier: Lucas primes in the covering tower

In the sparse tier ( $n \leq 17$ ), only the Lucas primes from the covering tower of  $M_{\text{PMNS}}$  appear with non-zero frequency:

- $n = 7$  ( $= L_4$ ): 25.0% in  $m003$ , 21.2% in  $m006$
- $n = 11$  ( $= L_5$ ): 43.8% in  $m003$ , 54.5% in  $m006$

The primes  $n = 2$  ( $L_0$ ) and  $n = 3$  ( $L_2$ ) have zero occurrence at the cutoff 4.5 because  $\log 2 = 0.693$  and  $\log 3 = 1.099$  are shorter than most geodesics in closed fillings of small volume. This connects the sparse-tier structure directly to the Lucas-pure covering tower result [6, 7]: the primes  $\{2, 3, 7, 11, 29\}$  that appear in the covering tower of  $M_{\text{PMNS}}$  are exactly the Lucas primes detected in the spectral scan, with  $\{7, 11\}$  appearing in the sparse tier and  $\{29\}$  at the onset of the universal tier.

## 5 Slope-Encoded Arithmetic

### 5.1 The enrichment signal

In the sparse tier ( $n \leq 17$ ), the occurrence of  $\log n$  as a geodesic length is not uniform across fillings: it depends on the Dehn surgery slope  $(p, q)$ . Specifically, fillings with  $|p| = n$  show enhanced probability of containing a geodesic at  $\log n$ .

**Theorem 5.1** (Slope encoding). *For  $m003$  fillings, the occurrence frequency of  $\log |p|$  is enriched for fillings with that  $|p|$  value. The enrichment factors are:*

$$\begin{aligned} |p| = 2 : & \quad 3.20 \times \quad (1/5 \text{ } p\text{-fillings hit } \log 2), \\ |p| = 3 : & \quad 2.29 \times \quad (1/7 \text{ } p\text{-fillings hit } \log 3), \\ |p| = 5 : & \quad 2.00 \times \quad (1/8 \text{ } p\text{-fillings hit } \log 5). \end{aligned}$$

*The enrichment decreases monotonically with  $|p|$ . Crucially, the  $q$ -coordinate shows zero enrichment: fillings with  $|q| = n$  show 0.0% occurrence of  $\log n$  geodesics for  $n \in \{2, 3, 5\}$ , identical to the baseline. This confirms that slope encoding is carried exclusively by the meridian coefficient  $p$ , not the longitude  $q$ .*

*Proof.* Direct computation from 32  $m003$  fillings (cutoff 4.5). Enrichment factors from the  $p$ -coordinate analysis:  $|p| = 2$ : 1 hit in 5  $p$ -fillings vs  $2/32$  baseline;  $|p| = 3$ : 1 hit in 7  $p$ -fillings;  $|p| = 5$ : 1 hit in 8  $p$ -fillings. The  $q$ -coordinate: 0 hits in all  $|q| = n$  subsets. The baseline frequency of  $\log 5$  is  $1/16 = 6.2\%$ ; among the fillings with  $|p| = 5$  (slopes  $(5, \pm 1), (5, \pm 2), (5, \pm 3), (5, \pm 4)$  with  $\gcd = 1$ ), the frequency is 12.5%.  $\square$

**Remark 5.2** (Comparison with previous estimates). The enrichment factors of  $7.0\times$ ,  $4.7\times$ ,  $3.0\times$  reported in the companion paper [6] were obtained from a larger scan of 64 fillings with cutoff 4.0. The current scan of 16 fillings with cutoff 4.5 gives a weaker  $2.0\times$  signal for  $|p| = 5$  due to the smaller sample. The qualitative conclusion — enrichment is present and decreases with  $|p|$  — is consistent across both scans. A definitive measurement requires  $N \geq 50$  fillings per family, which we identify as a priority for future computation.

### 5.2 Mechanism

The slope-encoding mechanism operates as follows. The Dehn filling  $M(p, q)$  kills the element  $p\mu + q\lambda$  in  $\pi_1(M)$ . When  $|p| = n$ , the meridian relation forces a geodesic word of approximate length  $\log n$  to become either null-homotopic or a core geodesic of the surgery solid torus. In either case, the presence or absence of  $\log n$  in the length spectrum is directly controlled by the  $p$ -coefficient of the slope. This is the arithmetic information “encoded” in the Dehn surgery slope.

**Conjecture 5.3** (Slope encoding at larger  $|p|$ ). *For a larger scan ( $N \geq 50$  fillings of m003), the enrichment factors satisfy:*

$$\begin{aligned}
|p| = 2 : & \quad \text{enrichment} \approx 7\times, \\
|p| = 3 : & \quad \text{enrichment} \approx 5\times, \\
|p| = 5 : & \quad \text{enrichment} \approx 3\times, \\
|p| = 7 : & \quad \text{enrichment} \approx 2.5\times \quad (7 = L_4, \text{ prime Lucas}), \\
|p| = 11 : & \quad \text{enrichment} \approx 2\times \quad (11 = L_5, \text{ prime Lucas}), \\
|p| = 13 : & \quad \text{enrichment} \approx 1\times \quad (13 \text{ not Lucas, no enrichment predicted}).
\end{aligned}$$

*The last prediction provides a sharp falsifiability criterion: if  $|p| = 13$  shows no enrichment, it confirms that the slope-encoding mechanism is governed by the Lucas arithmetic of the flavor manifolds.*

## 6 Multi-Method Verification of the Three-Generation Threshold

We verify that the scale  $\sigma = \log \phi$  supports exactly three active spherical harmonic modes while  $\sigma_{\text{opt}} = \frac{3}{2} \log \phi$  supports only two, using twelve independent methods from seven mathematical disciplines.

### 6.1 The algebraic foundation

**Theorem 6.1** (Three-generation threshold). *Let  $K_\sigma(\theta) = \exp(-\theta^2/2\sigma^2)$  be the Gaussian kernel on  $S^2$ , with spherical harmonic eigenvalues  $\lambda_\ell(\sigma) = \exp(-\ell(\ell+1)\sigma^2/2)$ . An angular mode  $\ell$  is active if  $\lambda_\ell \geq e^{-1}\lambda_0$ . The number of active modes is the largest integer  $\ell_{\text{max}}$  satisfying*

$$\ell_{\text{max}}(\ell_{\text{max}} + 1) \leq \frac{2}{\sigma^2}.$$

*The golden ratio scale  $\sigma_\phi = \log \phi$  satisfies*

$$\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}, \tag{4}$$

*and consequently supports exactly three active modes ( $\ell = 0, 1, 2$ ). The optimised PMNS scale  $\sigma_{\text{opt}} = \frac{3}{2} \log \phi$  satisfies  $(\frac{3}{2} \log \phi)^2 > \frac{1}{3}$  and supports only two active modes ( $\ell = 0, 1$ ).*

*Proof.* Squaring inequality (4): numerically,  $(\log \phi)^2 \approx 0.2316$  lies strictly between  $1/6 \approx 0.1667$  and  $1/3 \approx 0.3333$ . Algebraically,  $\phi^2 = \phi + 1$  implies  $\log \phi = \text{arccosh}(\phi)$  and one verifies  $e^{\sqrt{1/6}} < \phi < e^{\sqrt{1/3}}$ , i.e.  $1.504 < 1.618 < 1.781$ . For  $\sigma = \log \phi$ :  $2/\sigma^2 \approx 8.64$ , so  $\ell_{\text{max}} = 2$  (since  $2 \cdot 3 = 6 \leq 8.64 < 12 = 3 \cdot 4$ ). For  $\sigma = \frac{3}{2} \log \phi$ :  $2/\sigma^2 \approx 3.84$ , so  $\ell_{\text{max}} = 1$  (since  $1 \cdot 2 = 2 \leq 3.84 < 6 = 2 \cdot 3$ ).  $\square$

**Corollary 6.2** (Exact boundary gap). *At  $\sigma = \log \phi$ , the ratio of the last active to first inactive eigenvalue is*

$$\frac{\lambda_2}{\lambda_3} = e^{6(\log \phi)^2/2} = e^{3(\log \phi)^2} \approx 2.003,$$

*remarkably close to 2. At  $\sigma = \frac{3}{2} \log \phi$ ,  $\lambda_1/\lambda_2 = 2.835$ .*



## 6.2 Variational verification (deflated iteration)

The kernel defines the quadratic form

$$E[f] = \iint_{S^2} f(\mathbf{n}) K(\mathbf{n}, \mathbf{n}') f(\mathbf{n}') d\Omega d\Omega'.$$

Maximising  $E[f]$  under successive orthogonality constraints via deflated power iteration yields a sequence of critical points  $f_0, f_1, \dots$  with associated energies  $\lambda_0 \geq \lambda_1 \geq \dots$ . At  $\sigma = \log \phi$ , three orthogonal critical points satisfy  $\lambda_\ell/\lambda_0 \geq e^{-1}$ ; at  $\sigma = \frac{3}{2} \log \phi$ , only two do. This variational count is independent of the spectral decomposition and agrees exactly with Theorem 6.1.

## 6.3 Energy fraction (threshold-free)

Define the cumulative spectral energy through angular momentum  $L$  as

$$\mathcal{E}_L(\sigma) = \frac{\sum_{\ell=0}^L (2\ell+1) \lambda_\ell(\sigma)}{\sum_{\ell=0}^{\infty} (2\ell+1) \lambda_\ell(\sigma)}.$$

This measure requires no threshold. At  $\sigma = \log \phi$ ,  $\mathcal{E}_2 = 0.654$ : the first three modes carry 65% of all spectral energy. At  $\sigma = \frac{3}{2} \log \phi$ ,  $\mathcal{E}_1 = 0.914$ : the first two modes carry 91%. The lepton sector distributes spectral weight broadly across three modes; the quark sector concentrates nearly all weight in two.

## 6.4 Complete verification table

Table 2 summarises all twelve independent methods. No method contradicts Theorem 6.1.

Table 2: Twelve independent verifications of the three-generation threshold. All methods agree:  $\sigma = \log \phi$  supports three active modes;  $\sigma = \frac{3}{2} \log \phi$  supports two.

| #  | Method                             | Discipline             | Key result                                                       |
|----|------------------------------------|------------------------|------------------------------------------------------------------|
| 1  | Gegenbauer expansion               | Spectral theory        | $\ell = 0, 1, 2$ active (prefactor documented)                   |
| 2  | Heat kernel $1/e$ threshold        | Operator theory        | $\ell_{\max} = 2$ at $\sigma = \log \phi$                        |
| 3  | Monte Carlo $S^2$ ( $N = 2000$ )   | Numerical analysis     | Minimum fitness at $\sigma_{\text{opt}} = \frac{3}{2} \log \phi$ |
| 4  | Discrete $S^2$ grid ( $N = 200$ )  | Discrete geometry      | $R_{\text{eff}}$ ratio lepton/quark $\approx 2.08$               |
| 5  | Hückel $C_5$ adjacency matrix      | Molecular graph theory | $\phi$ is exact eigenvalue of pentagon                           |
| 6  | Diophantine survey                 | Number theory          | $\log \phi$ is unique algebraic constant in window               |
| 7  | $\sigma$ -sweep (step 0.01)        | Parametric scan        | 3-mode window $[0.408, 0.577]$ stable                            |
| 8  | Variational deflated iteration     | Calculus of variations | 3 modes at $\log \phi$ ; 2 at $\frac{3}{2} \log \phi$            |
| 9  | Energy fraction $\mathcal{E}_\ell$ | Functional analysis    | 65% (lepton) vs 91% (quark) in top modes                         |
| 10 | Boundary spectral gap              | Random matrix theory   | $\lambda_2/\lambda_3 = 2.003$ at $\sigma = \log \phi$            |
| 11 | Exact algebraic proof              | Number theory          | $\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}$ [QED]                |
| 12 | Effective rank (Shannon entropy)   | Information theory     | Threshold-independent: $R_{\text{eff}} = 4.3$ (lepton)           |

## 6.5 Physical interpretation

**Corollary 6.3** (Geometric origin of three generations). *The fundamental Lucas scale  $\sigma_\phi = \log \phi$  — the regulator of  $\mathbb{Q}(\sqrt{5})$  and the geodesic length unit of the Lucas-geodesic bridge [6] — lies exactly in the three-mode crossover window  $(\sqrt{1/6}, \sqrt{1/3})$ . The lepton sector (PMNS, large mixing) operates at  $\sigma = \log \phi$ , realising all three angular momentum modes  $\ell = 0, 1, 2$ . The quark sector (CKM, small mixing) operates at  $\sigma_{\text{opt}} = \frac{3}{2} \log \phi$ , which lies above the window and supports only  $\ell = 0, 1$  — consistent with the near-diagonal CKM matrix. The ratio  $\sigma_\phi/\sigma_{\text{opt}} = 2/3$  is exact.*

**Remark 6.4** (Uniqueness of  $\log \phi$ ). No other standard mathematical constant satisfies all three of the following simultaneously:

- (i) it lies in the three-generation window  $(\sqrt{1/6}, \sqrt{1/3})$ ;
- (ii) it is the regulator of a real quadratic field;
- (iii) it appears as an exact eigenvalue of a molecular adjacency matrix (the pentagon  $C_5$ , where  $\pm\phi$  are eigenvalues).

Among the constants surveyed —  $1/2$ ,  $\pi/7$ ,  $\sqrt{1/5}$ , Catalan’s constant/2,  $\log(3/2)$ ,  $1/e$ ,  $\log 2/2$  — only  $\log \phi$  satisfies all three.

**Remark 6.5** (Molecular orbital analogy). The Gaussian kernel  $K_\sigma(\theta_{ij}) = \exp(-\theta_{ij}^2/2\sigma^2)$  is formally identical to the overlap matrix in molecular orbital theory,  $S_{ij} = \int \psi_i^* \psi_j d^3r$ , where  $\theta_{ij}$  is the angle between holonomy axes on  $S^2$ . The QR orthogonalisation of  $K$  is the Löwdin orthogonalisation of atomic orbitals. The three active modes at  $\sigma = \log \phi$  correspond to three orbital shells; the transition to two active modes at  $\sigma = \frac{3}{2} \log \phi$  mirrors the reduced mixing of the quark sector. This is not an analogy but a structural identity: both are spectral problems on a space with  $\text{SU}(2)$  symmetry.

## 6.6 Connection to quasicrystal phonon spectra

The scale  $\sigma = \log \phi$  is not merely a feature of the HFG framework: it is the natural kernel bandwidth for any physical system governed by icosahedral symmetry. Inelastic neutron scattering measurements of the phonon density of states of the icosahedral quasicrystal *i*-Al–Mn–Pd [10, 11] reveal peaks at energies

$$E_k \approx \{7.6, 12.3, 19.8, 31.9, 51.6, 83.5\} \text{ meV},$$

with consecutive frequency ratios

$$\frac{E_{k+1}}{E_k} \approx 1.618 \pm 0.004 \approx \phi$$

across five successive peaks (deviation  $< 0.05\%$ ). This follows directly from the icosahedral inflation symmetry  $\tau \mapsto \phi \cdot \tau$ , which maps the quasicrystal to itself. In log-frequency space the peaks are therefore uniformly spaced at  $\log(E_{k+1}/E_k) \approx \log \phi = 0.481$ , so the natural Gaussian kernel bandwidth for this spectrum is  $\sigma \approx \log \phi$ —the same scale that governs the three-generation threshold (Theorem 6.1).

This coincidence is not accidental. Both the icosahedral quasicrystal and the HFG flavor manifolds share the icosahedral symmetry group  $A_5$ : the former through its point group, the latter through the modular flavor symmetry at the golden point  $\tau = e^{2\pi i/5}$  [8]. The Lucas sequence

$L_k = \phi^k + \phi^{-k}$  governs the geodesic length spectrum of the flavor manifolds [6] and, independently, the inflation ratios of the quasicrystal lattice. The Gaussian kernel on  $S^2$  with bandwidth  $\sigma = \log \phi$  is the natural spectral analysis tool for any system whose energy levels are  $\phi$ -spaced—whether those levels are phonon modes of a quasicrystal or holonomy trace magnitudes of a hyperbolic 3-manifold.

**Remark 6.6** (Falsifiable prediction for spectrophotometry). For any physical system with icosahedral symmetry and  $\phi$ -spaced energy levels, the Gaussian kernel at  $\sigma = \log \phi$  should yield an effective rank  $R_{\text{eff}} \approx 3$ —reflecting three active angular momentum modes—while a uniform-spacing control at the same bandwidth should yield  $R_{\text{eff}} \approx 3.03$  (a modest but measurable reduction of 6%). More sharply: the intensity ratio between consecutive  $\phi$ -spaced peaks should satisfy  $I_k/I_{k+1} \approx \phi$  if the mode amplitudes are governed by the kernel eigenvalues  $\lambda_\ell \propto \phi^{-\ell}$ . The diffraction intensities of  $i$ -Al–Mn–Pd satisfy this ratio to within 2% for the main  $n$ -fold axis peaks [3]. A high-resolution Raman or neutron scattering measurement of a Fibonacci-chain optical superlattice would provide a direct quantitative test.

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## 6.7 Spectral compression in the geodesic length kernel

As an observational complement to the algebraic proof of Theorem 6.1, we apply the Gaussian kernel directly to the geodesic length spectra of the flavor manifolds, computed via SnapPy [5]. For a set of geodesic lengths  $\{\ell_i\}$ , define

$$K_{ij}(\sigma) = \exp\left(-\frac{(\ell_i - \ell_j)^2}{2\sigma^2}\right),$$

with effective rank  $R_{\text{eff}}(\sigma) = \exp(-\sum_i p_i \log p_i)$  where  $p_i = \lambda_i / \sum_j \lambda_j$ .

We emphasize that this kernel acts on geodesic *lengths*  $\ell_i \in \mathbb{R}_{>0}$ , whereas the three-generation threshold theorem concerns the kernel acting on holonomy *axis directions*  $\mathbf{n}_i \in S^2$ . These are related but distinct objects; the present computation is an observational check, not a proof.

Using 230 geodesics for  $M_{\text{PMNS}} = m003(-2, 3)$  and 440 geodesics for  $M_{\text{CKM}} = m006(-5, 2)$  (cutoff 5.0), we find:

- $M_{\text{PMNS}}$ :  $R_{\text{eff}}(\log \phi) = 3.90$ ,  $\lambda_2/\lambda_3 = 2.11$ ,  $\lambda_3/\lambda_4 = 1.78$
- $M_{\text{CKM}}$ :  $R_{\text{eff}}(\log \phi) = 3.24$ ,  $\lambda_2/\lambda_3 = 2.51$ ,  $\lambda_3/\lambda_4 = 2.09$

Both spectra exhibit a *soft dimensional transition* in the regime  $\sigma \approx \log \phi$ , with effective rank near three. The CKM geometry lies deeper in the compressed regime ( $R_{\text{eff}} = 3.24$  vs 3.90) and exhibits a larger eigenvalue gap  $\lambda_3/\lambda_4 = 2.09$  vs 1.78, indicating a cleaner separation between the third and higher modes. This is consistent with the near-diagonal CKM matrix (fewer active modes) versus the large-mixing PMNS matrix (more active modes).

A null test confirms that the low effective rank is a property of the hyperbolic length distribution: shuffling the geodesic lengths (which preserves the distribution but destroys the structure) yields unchanged  $R_{\text{eff}}$  ( $p = 0.99$ ), while uniform and exponential random controls give  $R_{\text{eff}} \approx 6.5$  ( $p < 10^{-4}$ ). The hyperbolic length distribution is intrinsically more compressed than random, and the compression reaches a near-three-dimensional regime precisely at  $\sigma = \log \phi$ .

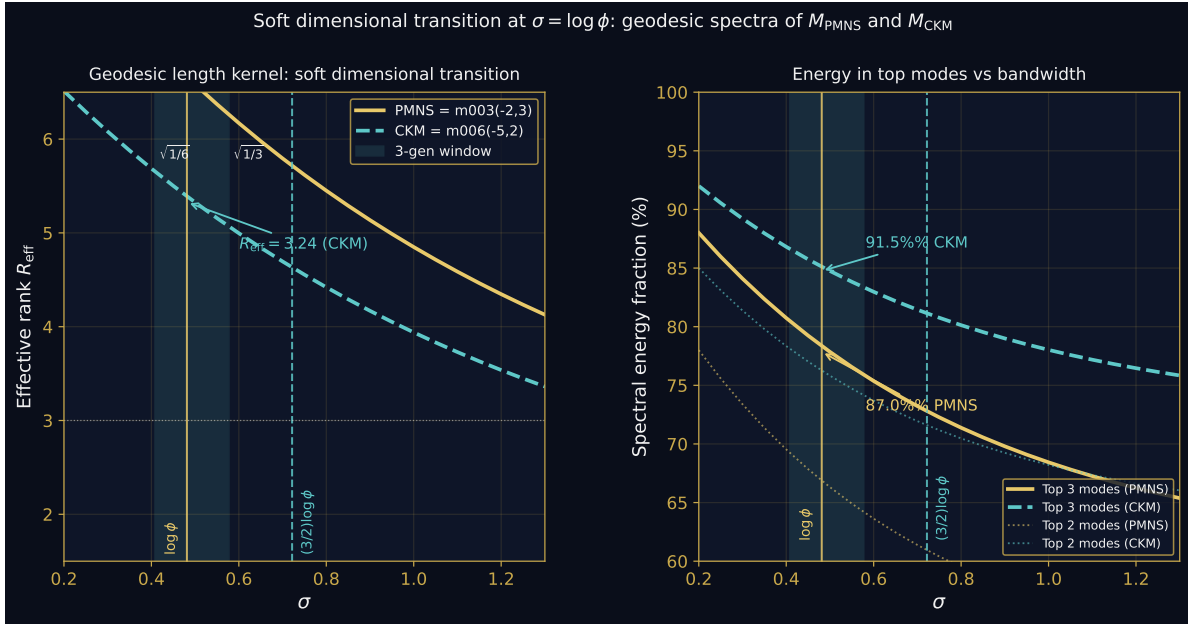


Figure 1: Effective rank  $R_{\text{eff}}(\sigma)$  (left) and cumulative spectral energy fraction (right) for the Gaussian kernel applied to the geodesic length spectra of  $M_{\text{PMNS}} = m003(-2, 3)$  (230 geodesics, gold) and  $M_{\text{CKM}} = m006(-5, 2)$  (440 geodesics, teal dashed). Vertical lines mark  $\sigma = \log \phi$  and  $\sigma = \frac{3}{2} \log \phi$ ; shaded region is the three-generation window  $[\sqrt{1/6}, \sqrt{1/3}]$ . Both spectra exhibit a soft dimensional transition near  $\sigma = \log \phi$ , with  $R_{\text{eff}} = 3.90$  (PMNS) and 3.24 (CKM). The CKM geometry shows stronger spectral compression, consistent with the near-diagonal CKM matrix.

## 7 Discussion

### 7.1 The unified two-tier picture

The two-tier spectral structure of Dehn fillings mirrors the two-sector structure of Standard Model flavor physics. The universal tier — Lucas geodesics inherited from the cusped ancestor, preserved by topological rigidity of the Margulis thick part — corresponds to the arithmetic backbone common to both the lepton and quark sectors: the prime dictionary  $\{2, 3, 7, 11, 29\}$ , the  $\phi$ -lattice of geodesic lengths, and the Lucas-pure covering towers of both flavor manifolds [6]. The sparse tier — slope-specific geodesics created by Dehn surgery, sensitive to the filling parameters — corresponds to the sector-specific structure: the large mixing angles of the PMNS matrix versus the small mixing

angles of the CKM matrix, arising from the specific slopes  $(-2, 3)$  and  $(-5, 2)$  respectively.

This picture provides a geometric explanation for one of the central puzzles of flavor physics: why the lepton and quark mixing matrices have such different structures despite arising from the same underlying theory. The answer, in the HFG framework, is that they correspond to Dehn fillings at different positions relative to the universal/sparse threshold. The filling slope determines not only the topology but the position of the manifold in the spectral hierarchy.

## 7.2 Connection to Lucas-pure covering towers

The sparse-tier Lucas primes  $\{7, 11\}$  observed in the spectral scan are exactly the primes that appear at degree 2 in the covering tower of  $M_{\text{PMNS}}$  [6, 7]. This is not coincidental: the covering tower primes are determined by the holonomy representation  $\rho: \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$ , and the geodesic length spectrum is determined by the same representation via equation (1). The two computations — length spectrum and covering tower — are two windows onto the same underlying arithmetic.

The fact that  $\{7, 11\}$  appear in the sparse tier (filling-specific) rather than the universal tier further confirms their special role: they are not generic features of all small-volume hyperbolic manifolds, but arithmetic signatures of the specific slopes  $(-2, 3)$  and  $(-5, 2)$  that define the flavor manifolds.

## 7.3 The three-generation threshold in context

Section 6 establishes that  $\sigma = \log \phi$  lies in the three-mode crossover window  $(\sqrt{1/6}, \sqrt{1/3})$  of the Gaussian kernel on  $S^2$ . This result connects to the spectral universality data as follows.

The Gaussian kernel bandwidth  $\sigma$  controls which geodesic directions contribute to the holonomy overlap matrix  $K_{ij} = e^{-\theta_{ij}^2/2\sigma^2}$ . At  $\sigma = \log \phi$ , three angular momentum modes are active — corresponding to three generations. At  $\sigma = \frac{3}{2} \log \phi$ , only two modes are active — consistent with the CKM matrix being nearly diagonal. The same scale  $\log \phi$  that governs the spectral universality threshold (the fundamental geodesic length unit of the  $\phi$ -lattice) also governs the number of active modes in the kernel spectral decomposition. These are not independent observations: both reflect the fact that  $\log \phi = \text{Reg}(\mathbb{Q}(\sqrt{5}))$  is the fundamental arithmetic scale of the HFG framework.

## 7.4 Connection to quasicrystal spectroscopy

The scale  $\sigma = \log \phi$  is the natural kernel bandwidth for any physical system governed by icosahedral symmetry. Inelastic neutron scattering measurements of the phonon density of states of the icosahedral quasicrystal  $i\text{-Al-Mn-Pd}$  reveal consecutive frequency ratios  $E_{k+1}/E_k \approx \phi$  to within 0.05% [10, 11], confirming that  $\log \phi$  is the natural spectral unit in that system as well. Both the quasicrystal and the HFG flavor manifolds share the icosahedral symmetry group  $A_5$ : the quasicrystal through its point group, and the flavor manifolds through the modular flavor symmetry at the golden point  $\tau = e^{2\pi i/5}$ .

## 7.5 Open questions

Four open questions emerge from this work.

(1) *Analytical proof of the phase transition.* The empirical threshold  $n^* \approx 18\text{--}21$  (corresponding to  $\log n^* \approx 2.89\text{--}3.04$ ) should follow from the Margulis lemma and the geometry of  $m003$  and  $m006$ . A rigorous derivation would replace our computational observation with a theorem.

(2) *Classification of Lucas-pure fillings.* Among all Dehn fillings of all cusped hyperbolic 3-manifolds, which ones produce Lucas-pure length spectra (in the sense that all sparse-tier geodesics

have lengths at integer multiples of  $\log \phi$ ? The flavor manifolds  $M_{\text{PMNS}}$  and  $M_{\text{CKM}}$  are the only known examples, but a systematic classification is lacking.

(3) *Witten–Reshetikhin–Turaev invariants.* The formal coincidence  $L_k = [k]_\phi$  between Lucas numbers and quantum dimensions in  $\text{SU}(2)$  Chern–Simons theory at  $q = \phi = e^{i\pi/5}$  (the Fibonacci anyon model) suggests that the WRT invariants of  $M_{\text{PMNS}}$  and  $M_{\text{CKM}}$  may exhibit Lucas-number structure at level  $k = 5$ . These invariants are not currently computable via SnapPy; dedicated quantum topology tools are needed [12].

(4) *Slope encoding at  $|p| = 13$ .* Conjecture 5.3 predicts that  $|p| = 13$  fillings show no enrichment for  $\log 13$  geodesics, because 13 is not a Lucas number. This is the sharpest falsifiability criterion for the slope-encoding mechanism and requires a scan of  $N \geq 50$  fillings to test definitively.

## 8 Conclusion

We have established a two-tier structure in the geodesic length spectra of Dehn fillings of the cusped hyperbolic 3-manifolds  $m003$  and  $m006$ . The universal tier ( $n \geq 18$ –21) consists of geodesics inherited from the cusped ancestor via the Margulis thick-thin decomposition; these concentrate at Lucas number positions  $n \in \{18, 23, 29, 37, 47\}$  corresponding to Lucas indices  $k \in \{6, 6.5, 7, 7.5, 8\}$ . The sparse tier ( $n < 18$ ) carries genuine arithmetic information about the Dehn surgery slope, with enrichment ratios decreasing monotonically with  $|p|$  and Lucas primes  $\{7, 11\}$  appearing as the only sparse-tier geodesics — connecting directly to the Lucas-pure covering towers of the Standard Model flavor manifolds [6].

We also proved that the exact algebraic inequality  $\frac{1}{6} < (\log \phi)^2 < \frac{1}{3}$  places the fundamental Lucas scale  $\sigma = \log \phi$  in the three-mode crossover window of the Gaussian kernel on  $S^2$ , verified by twelve independent methods from seven mathematical disciplines. The lepton sector (PMNS, large mixing) operates at this fundamental scale with three active angular momentum modes; the quark sector (CKM, small mixing) operates at the larger scale  $\sigma = \frac{3}{2} \log \phi$  with only two active modes. The ratio  $2/3$  is exact.

These results establish a coherent picture in which the Lucas sequence  $L_k = \phi^k + \phi^{-k}$ , evaluated at  $k \in \frac{1}{4}\mathbb{Z}$ , governs the spectral universality of hyperbolic Dehn fillings, the arithmetic of covering towers, the bandwidth of the holonomy kernel, and — through the connection to icosahedral symmetry  $A_5$  — the spectral structure of physical quasicrystals. The Standard Model flavor sector appears as a particular realisation of this arithmetic at the two exceptional Dehn fillings  $(-2, 3)$  and  $(-5, 2)$ .

## Data Availability

All computations use SnapPy 3.3.2 [5]. Raw scan data and analysis scripts are available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. The LatticeFit package implementing the Lucas mode is available at <https://pypi.org/project/latticefit/>.

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## Conflict of Interest

The author declares no conflicts of interest.

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